

ANSWER SCHEME SET 6

SECTION A		
1.	(a)	(i) $\lim_{x \rightarrow 2} \frac{x-2}{x^4-16}$ $= \lim_{x \rightarrow 2} \frac{x-2}{(x-2)(x+2)(x^2+4)}$ $= \lim_{x \rightarrow 2} \frac{1}{(x+2)(x^2+4)}$ $= \frac{1}{(2+2)(2^2+4)}$ $= \frac{1}{32}$ (ii) $\lim_{x \rightarrow \infty} \frac{(2-x)(x-1)}{(x-3)^2}$ $\lim_{x \rightarrow \infty} \frac{(2-x)(x-1)}{(x-3)^2}$ $= \lim_{x \rightarrow \infty} \frac{3x - x^2 - 2}{x^2 - 6x + 9}$ $= \lim_{x \rightarrow \infty} \frac{\frac{3x}{x^2} - \frac{x^2}{x^2} - \frac{2}{x^2}}{\frac{x^2}{x^2} - \frac{6x}{x^2} + \frac{9}{x^2}}$ $= \lim_{x \rightarrow \infty} \frac{\frac{3}{x} - 1 - \frac{2}{x^2}}{1 - \frac{6}{x} + \frac{9}{x^2}}$ $= \frac{0-1-0}{1-0+0}$ $= -1$
	(b)	$\lim_{x \rightarrow \infty} \frac{1}{2-\sqrt{x}} = 0$ Thus, $y = 0$ is horizontal asymptote.
2.	(a)	$x = t^2 \Rightarrow \frac{dx}{dt} = 2t$ $y = t^3 - 3t \Rightarrow \frac{dy}{dt} = 3t^2 - 3$ $\frac{dy}{dx} = \frac{3t^2 - 3}{2t}$

		$\left. \frac{dy}{dx} \right _{t=\sqrt{3}} = \frac{3(\sqrt{3})^2 - 3}{2(\sqrt{3})} = \sqrt{3}$												
	(b)	$y = 3^{2x-5}$ $\frac{dy}{dx} = 3^{2x-5} \cdot 2 \cdot \ln 3 = 2 \ln 3 \left(3^{2x-5} \right)$ $\frac{d^2y}{dx^2} = 2 \ln 3 \left[3^{2x-5} \cdot 2 \cdot \ln 3 \right] = (2 \ln 3)^2 3^{2x-5}$ $\left. \frac{d^2y}{dx^2} \right _{x=3} = (2 \ln 3)^2 3^{2(3)-5} = 12(\ln 3)^2$												
3.		$f(x) = x^3 - 6x^2 + 9x - 3$ $f'(x) = 3x^2 - 12x + 9 = 0$ $x^2 - 4x + 3 = 0$ $(x-3)(x-1) = 0 \Rightarrow x = 3; x = 1$ Critical points are (1,1) and (3,-3). <table><tr><td>Intervals</td><td>(0,1)</td><td>(1,3)</td><td>(3,5)</td></tr><tr><td>Sign of $f'(x)$</td><td>+ ve</td><td>- ve</td><td>+ ve</td></tr><tr><td>Conclusion</td><td></td><td></td><td></td></tr></table> Minimum point : (3,-3) and maximum point : (1,1) .	Intervals	(0,1)	(1,3)	(3,5)	Sign of $f'(x)$	+ ve	- ve	+ ve	Conclusion			
Intervals	(0,1)	(1,3)	(3,5)											
Sign of $f'(x)$	+ ve	- ve	+ ve											
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SECTION B														
1.		$\frac{p+qi}{3i} = (3+\sqrt{-16}-i^3)$ $\left(\frac{p+qi}{3i} \right) \times \left(\frac{-3i}{-3i} \right) = (3+4i-i^2i)$ $\frac{q}{3} - \frac{p}{3}i = 3+5i$ $\frac{q}{3} = 3 \Rightarrow q = 9, \quad \frac{-p}{3} = 5 \Rightarrow p = -15$												

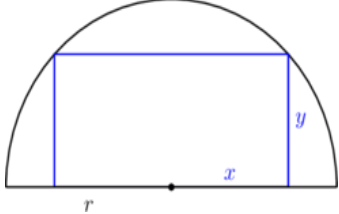
2.	(a)	$\log_2 x - \log_4(3x + 4) = 0$ $\log_2 x = \frac{\log_2(3x + 4)}{\log_2 4}$ $\log_2 x = \frac{\log_2(3x + 4)}{2}$ $\log_2 x^2 = \log_2(3x + 4)$ $x^2 - 3x - 4 = 0$ $(x - 4)(x + 1) = 0$ $x = 4 \quad , x = -1$ $x = 4 \quad \text{only (accepted)}$																									
	(b)	$\frac{-4 - x}{x - 3} \geq x + 4$ $\frac{-4 - x}{x - 3} - (x + 4) \geq 0$ $\frac{-4 - x}{x - 3} - \frac{(x + 4)(x - 3)}{x - 3} \geq 0$ $\frac{-x^2 - 2x + 8}{x - 3} \geq 0$ $\frac{x^2 + 2x - 8}{x - 3} \leq 0$ $\frac{(x - 2)(x + 4)}{x - 3} \leq 0$ applying sign table <table><tr><td>Intervals</td><td>$(-\infty, -4)$</td><td>$(-4, 2)$</td><td>$(2, 3)$</td><td>$(3, \infty)$</td></tr><tr><td>$x - 2$</td><td>-</td><td>-</td><td>+</td><td>+</td></tr><tr><td>$x + 4$</td><td>-</td><td>+</td><td>+</td><td>+</td></tr><tr><td>$x - 3$</td><td>-</td><td>-</td><td>-</td><td>+</td></tr><tr><td>$\frac{(x - 2)(x + 4)}{x - 3}$</td><td>-</td><td>+</td><td>-</td><td>+</td></tr></table> $\{x : x \leq -4 \cup 2 \leq x < 3\}$	Intervals	$(-\infty, -4)$	$(-4, 2)$	$(2, 3)$	$(3, \infty)$	$x - 2$	-	-	+	+	$x + 4$	-	+	+	+	$x - 3$	-	-	-	+	$\frac{(x - 2)(x + 4)}{x - 3}$	-	+	-	+
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$\frac{(x - 2)(x + 4)}{x - 3}$	-	+	-	+																							

3.	(a)	$f(x) = \frac{x^2 + 1}{5}, \geq 0$ $ff^{-1}(x) = x$ $\frac{(f^{-1}(x))^2 + 1}{5} = x$ $f^{-1}(x) = \sqrt{5x - 1}$ $\text{hence, } f(g(x)) = \frac{1}{5}(e^{2(3x-1)} + 1)$ $f^{-1}f(g(x)) = g(x)$ $g(x) = \sqrt{5\left(\frac{1}{5}(e^{2(3x-1)} + 1)\right)} - 1$ $= e^{(3x-1)}$
	(b)	$g(f(2)) = g\left(\frac{2^2 + 1}{5}\right)$ $= e^{(3(1)-1)}$ $= e^2$ ≈ 7.389
4.	(a)	$(f \circ g)(x) = f(g(x)) = f(x - 3) = e^{ x-3 }$ $\rightarrow (f \circ g)(x) = \begin{cases} e^{x-3} & , x \geq 3 \\ e^{-(x-3)} & , x < 3 \end{cases}$
	(b)	$\text{let } m = (f \circ g)(x)$ $mm^{-1}(x) = x$ $e^{m^{-1}(x)-3} = x$ $m^{-1}(x) = \ln x + 3$ $(f \circ g)^{-1}(x) = \ln x + 3$

	(c)	$h \circ f(x) = \frac{2e^x}{1-3e^x}$ $h(e^x) = \frac{2e^x}{1-3e^x}$ $\text{let } a = e^x$ $h(a) = \frac{2a}{1-3a}$ $\therefore h(x) = \frac{2x}{1-3x}$ $h(x_1) = h(x_2)$ $\frac{2x_1}{1-3x_1} = \frac{2x_2}{1-3x_2}$ $2x_1(1-3x_2) = 2x_2(1-3x_1)$ $x_1 - 3x_1x_2 = x_2 - 3x_1x_2$ $\therefore x_1 = x_2$ Thus, $h(x)$ is one to one function.
5.	(a)	$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \left(\frac{-2x}{x} + 5x \right) = -2$ $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \left(\frac{2x}{x} + 5x \right) = 2$
	(b)	$g(x)$ is continuous at a $\Leftrightarrow$ $\lim_{x \rightarrow a} g(x) = g(a)$ $\lim_{x \rightarrow a} \sqrt{5-x} = 3a-1$ $\sqrt{5-a} = 3a-1$ $5-a = 9a^2 - 6a + 1$ $9a^2 - 5a - 4 = 0$ $(9a+4)(a-1) = 0$ $a = -\frac{4}{9}, a = 1$

		at $a = -\frac{4}{9}$ $\lim_{x \rightarrow a^-} g(x) = \sqrt{5 + \frac{4}{9}} = \frac{7}{3}$ $\lim_{x \rightarrow a^+} g(x) = 3\left(-\frac{4}{9}\right) - 1 = -\frac{7}{3}$ $\lim_{x \rightarrow a^-} g(x) \neq \lim_{x \rightarrow a^+} g(x)$ $\rightarrow \lim_{x \rightarrow a} g(x) \text{ does not exist}$ $\therefore g(x)$ is not continuous at $a = -\frac{4}{9}$ $a = 1$ $\lim_{x \rightarrow a^-} g(x) = \sqrt{5-1} = 2$ $\lim_{x \rightarrow a^+} g(x) = 3(1) - 2 = 2$ $\lim_{x \rightarrow a^-} g(x) = \lim_{x \rightarrow a^+} g(x)$ $\rightarrow \lim_{x \rightarrow a} g(x) = 2$ $g(1) = 3(1) - 1 = 2$ $\lim_{x \rightarrow a} g(x) = g(1)$ $\therefore g(x)$ is continuous at $a = 1$
	(c)	(i) VA : $x^2 - 4 = 0 \Rightarrow x = \pm 2$ $p = -2, q = 2$ (ii) HA $\lim_{x \rightarrow \pm\infty} \frac{2x}{x^2 - 4}$ $= \lim_{x \rightarrow \pm\infty} \frac{\frac{2x}{x^2}}{\frac{x^2 - 4}{x^2}}$ $= \lim_{x \rightarrow \pm\infty} \frac{\frac{2}{x}}{1 - \frac{4}{x^2}}$ $= 0$ HA $y = 0$

	6	show that $\frac{d}{dx}(\tan x) = \sec^2 x$ by using $\tan x$ in terms of $\sin x$ and $\cos x$: $\frac{d}{dx}(\tan x)$ $\frac{d}{dx}\left(\frac{\sin x}{\cos x}\right)$ use quotient rule: $= \frac{\cos x \cos x - \sin x(-\sin x)}{\cos^2 x}$ $= \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$ $= \frac{1}{\cos^2 x}$ $= \sec^2 x$ shown
6.	(a)	$\frac{d^2 y}{dx^2} = 2 \sec x (\sec x \tan x)$ $= 2 \sec^2 x \tan x$ $= 2(1 + \tan^2 x) \tan x$ $= 2(1 + y^2)y$ $\frac{d^2 y}{dx^2} > 0$ $2(1 + y^2)y > 0$ since, $(1 + y^2) > 0, y > 0 \rightarrow \tan x > 0$ $\therefore 0 < x < \frac{\pi}{2}$
	(b)	$y = \tan(x + y)$ $\frac{dy}{dx} = \sec^2(x + y) \left(1 + \frac{dy}{dx}\right)$ $\frac{dy}{dx} = \sec^2(x + y) + \sec^2(x + y) \frac{dy}{dx}$ $\frac{dy}{dx} - \sec^2(x + y) \frac{dy}{dx} = \sec^2(x + y)$ $\left(1 - \sec^2(x + y)\right) \frac{dy}{dx} = \sec^2(x + y)$ $\frac{dy}{dx} = \frac{\sec^2(x + y)}{1 - \sec^2(x + y)}$

		$x = y = \alpha$ $\frac{dy}{dx} = \frac{\sec^2(2\alpha)}{1 - \sec^2(2\alpha)}$ $= \sec^2(2\alpha) \cdot \frac{1}{1 - \sec^2(2\alpha)}$ $= \frac{1}{\cos^2(2\alpha)} \cdot \frac{1}{-\tan^2(2\alpha)}$ $= \frac{1}{\cos^2(2\alpha)} \cdot -\frac{\cos^2(2\alpha)}{\sin^2(2\alpha)}$ $= -\frac{1}{\sin^2(2\alpha)}$ $= -\operatorname{cosec}^2(2\alpha)$
7.	(a)	 $x^2 + y^2 = 100$ $y = \sqrt{100 - x^2}$ area rectangle, $A = 2xy = 2x\sqrt{100 - x^2}$ $A' = 2\sqrt{100 - x^2} - \frac{2x^2}{\sqrt{100 - x^2}}$ $0 = 2\sqrt{100 - x^2} - \frac{2x^2}{\sqrt{100 - x^2}}$ $2\sqrt{100 - x^2} = \frac{2x^2}{\sqrt{100 - x^2}}$ $\Rightarrow x = \sqrt{50}, y = \sqrt{50}$ Area max, $A = 2\sqrt{50}(\sqrt{100 - (\sqrt{50})^2}) = 100 \text{ cm}^2$

	(b)	$\frac{dx}{dt} = 2.5 \text{ cm} / \text{s}$ $V = x^3$ $\frac{dV}{dx} = 3x^2$ $\frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dt}$ $\frac{dV}{dt} = 3x^2 \times 2.5 = 7.5x^2$ $\text{when } x = 8 \text{ cm}, \frac{dV}{dt} = 7.5(8)^2$ $= 480 \text{ cm}^3 / \text{s}$
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